Deep Learning Roadmap

Zero se job-ready tak, phase by phase

Duration: 10–14 mahine **Effort:** 15–20 ghante / hafta **Level:** Beginner se Advanced

Shuru karne se pehle: raasta chuno

Do alag alag raaste hain aur log inko hamesha confuse karte hain. Ghalat raasta chuna to 6 se 12 mahine zaya honge.

Raasta	Kya seekhoge	Kis ke liye
1. Deep Learning Engineer	Models samajhna, train karna, architectures design karna, research papers padhna	Research, serious ML engineering, proper foundation. 10–14 mahine
2. AI App Developer	LLM APIs, RAG, agents, chatbots, product banana	Jaldi freelancing ya job. 2–3 mahine

Honest baat: Agar goal sirf jaldi kamana hai to seedha raasta 2 lo. Pura deep learning roadmap tumhare liye waqt ka zaya hoga. Pakistan ka freelancing aur job market abhi AI app development mein zyada garam hai. Ye document raasta 1 ke liye hai, yani proper foundation banane walon ke liye.

Poora roadmap ek nazar mein

Phase	Topic	Duration	Output
0	Python aur tools	3–4 hafte	Data cleaning + plots project
1	Math foundations	4–6 hafte (parallel)	Gradient descent intuition
2	Classical Machine Learning	5–6 hafte	Kaggle submission
3	Neural Network fundamentals	6–8 hafte	NumPy se scratch NN
4	PyTorch aur Computer Vision	6–8 hafte	Custom image classifier
5	Sequences aur Transformers	8 hafte	Scratch se chhota GPT
6	LLMs aur Generative Models	8 hafte	LoRA fine-tune + RAG app
7	Deployment aur MLOps	4–6 hafte	Live deployed model

Phase 0 · Python aur Tools

3–4 HAFTE

Sirf Python ki basics kaafi nahi hain. Ye specific cheezein chahiye:

- **Core Python:** functions, classes, list aur dict comprehensions, decorators ka basic idea
- **NumPy** achi tarah. Array indexing, broadcasting, vectorization
- **Pandas** data loading aur cleaning ke liye
- **Matplotlib** basic plotting ke liye
- **Git aur GitHub.** Har project push karo, yehi tumhara portfolio hai
- **Google Colab ya Kaggle Notebooks.** Free GPU milta hai, apna GPU abhi lene ki zarurat nahi

NumPy skip mat karna. Log ise boring samajh kar chhod dete hain, phir Phase 3 mein backpropagation samajhne mein atak jate hain. Neural network internally NumPy hi hai.

Project: Kaggle se koi bhi CSV dataset lo. Cleaning karo, 5 meaningful plots banao, aur 3 insights likho.

Phase 1 · Math Foundations

4–6 HAFTE · PARALLEL CHALAO

Ye phase blocking nahi hona chahiye. Math seekhte raho lekin sath sath code bhi karo, warna teen mahine sirf math mein nikal jayenge aur motivation khatam ho jayegi.

LINEAR ALGEBRA

- Vectors aur matrices, matrix multiplication
- Dot product, transpose, inverse
- Eigenvalues aur eigenvectors ka basic idea

CALCULUS

- Derivatives aur partial derivatives
- **Chain rule** (sabse important, backpropagation isi pe khara hai)
- Gradient ka matlab aur geometric intuition

PROBABILITY AUR STATISTICS

- Distributions, mean, variance, standard deviation
- Bayes rule, conditional probability, expectation

RESOURCES

- **3Blue1Brown** – "Essence of Linear Algebra" aur "Essence of Calculus" (YouTube). Intuition ke liye best material hai
- **Khan Academy** – agar practice problems chahiye
- **Mathematics for Machine Learning** – Imperial College, Coursera. Structured course chahiye to

> **Proofs rattne ki zarurat nahi.** Tumhe samajhna hai ke gradient ka matlab kya hai aur wo kaam kaise karta hai, theorems derive karna nahi.

Phase 2 · Classical Machine Learning

5–6 HAFTE

Log ye phase skip kar ke seedha neural networks pe kood jate hain. Ye ghalti hai. Overfitting, regularization, train/test split, evaluation metrics, bias-variance tradeoff, ye sab yahan sikhte ho. Deep learning mein wahi concepts dubara aate hain, bas bade scale pe.

ALGORITHMS

- Linear Regression aur Logistic Regression
- Decision Trees, Random Forest, Gradient Boosting (XGBoost)
- k-Nearest Neighbors, SVM
- k-Means clustering, PCA

CONCEPTS JO INTERVIEW MEIN POOCHHE JATE HAIN

- Cross-validation aur train/validation/test split
- Confusion matrix, precision, recall, F1 score, ROC-AUC
- Bias-variance tradeoff
- Feature engineering aur **data leakage**
- Imbalanced datasets ko handle karna

RESOURCES

- **Machine Learning Specialization** – Andrew Ng, Coursera. Audit free hai
- **Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow** – Aurélien Géron. Agar sirf ek kitaab leni ho to yehi lo
- **Scikit-learn** official documentation aur user guide

> **Project:** Titanic ya House Prices pe Kaggle submission karo. Phir koi aisa dataset lo jo tumhe genuinely interest ho, jaise cricket stats ya weather data.

Phase 3 · Neural Network Fundamentals

6–8 HAFTE · SABSE AHAM PHASE

Ye poore roadmap ka sabse important phase hai. Yahan jaldi bilkul mat karna. Agar yahan solid ho gaye to aage sab kuch aasan lagega.

CORE CONCEPTS

- Perceptron aur multi-layer networks
- Activation functions: ReLU, sigmoid, tanh, softmax (aur kab kaun sa use karna hai)
- Forward pass aur loss functions: MSE, cross-entropy
- **Backpropagation** haath se, kaghaz pe. Chhote network pe khud derive karo
- Gradient descent variants: SGD, momentum, RMSprop, Adam
- Vanishing aur exploding gradients

OVERFITTING KA ILAJ

- Dropout, weight decay (L2), early stopping
- Data augmentation
- Batch normalization aur layer normalization
- Weight initialization (Xavier, He) aur ye matter kyun karta hai

HYPERPARAMETER TUNING

- Learning rate sabse zyada matter karta hai, isi pe waqt lagao
- Batch size, epochs, network depth aur width
- Learning rate schedules aur warmup

> **Is phase ka sabse zaroori kaam:** NumPy mein scratch se ek neural network likho. Sirf NumPy, koi framework nahi. Do se teen layers, MNIST pe train karo, backprop khud implement karo. Jab tak ye khud nahi likha, tum samjhe nahi ho. Ye ek kaam tumhe 80% logon se aage kar dega.

RESOURCES

- **Andrej Karpathy – "Neural Networks: Zero to Hero"** (YouTube). Free hai aur is topic pe best material hai jo exist karta hai. Micrograd wali video se shuru karo
- **Deep Learning Specialization** – Andrew Ng, agar structured course chahiye
- **d2l.ai** (Dive into Deep Learning) – free online book, har concept ke sath code

Phase 4 · PyTorch aur Computer Vision

6–8 HAFTE

> **PyTorch seekho, TensorFlow pe waqt zaya mat karo.** Research, open source community, aur ab zyadatar industry PyTorch pe shift ho chuki hai. TensorFlow sirf tab seekhna jab kisi specific job ki requirement ho.

PYTORCH

- Tensors aur autograd
- `nn.Module`, custom layers, `DataLoader` aur `Dataset`
- Training loop khud likhna (shortcut libraries baad mein)
- GPU pe move karna, mixed precision training
- Model save aur load karna, checkpointing

CONVOLUTIONAL NEURAL NETWORKS

- Convolution operation, kernels, stride, padding
- Pooling aur receptive field
- Architectures: LeNet, AlexNet, VGG, **ResNet**
- Residual connections kyun kaam karte hain (ye zaroor samjho)

PRACTICAL SKILLS

- **Transfer learning aur fine-tuning.** Real life mein 90% cases mein tum yehi karoge, scratch se train nahi
- Data augmentation pipelines

- Object detection ka basic idea (YOLO family)
- Image segmentation (U-Net)

RESOURCES

- **PyTorch** official tutorials, especially "Deep Learning with PyTorch: A 60 Minute Blitz"
- **fast.ai** – "Practical Deep Learning for Coders". Top-down approach hai, pehle chalao phir samjho
- **Stanford CS231n** lectures aur course notes. Notes khud hi ek kitaab ke barabar hain

> **Project:** Apne khud ke collect kiye hue photos pe image classifier banao. Ready dataset se zyada seekhne ko milta hai kyunke messy data ka saamna karna padta hai. Phir ek pretrained ResNet fine-tune karo aur dono ka comparison likho.

Phase 5 · Sequences aur Transformers

8 HAFTE

SEQUENCE MODELS

- RNN, LSTM, GRU. Inko samjho lekin zyada waqt mat do, ab ye mostly legacy hain
- Sequence-to-sequence aur encoder-decoder idea
- Word embeddings: Word2Vec, GloVe

TRANSFORMERS (YAHAN RUK KAR POORI TARAH SAMJHO)

- **Attention mechanism** – query, key, value
- Self-attention aur multi-head attention
- Positional encoding
- Poora Transformer block: attention + feedforward + residual + layer norm
- Tokenization aur BPE
- BERT vs GPT: encoder vs decoder, masked LM vs causal LM
- Pretraining aur fine-tuning ka paradigm

RESOURCES

- **"Attention Is All You Need"** – original paper. Pehli baar samajh na aaye to normal hai
- **"The Illustrated Transformer"** – Jay Alammar ka blog. Paper se pehle ye padho
- **Andrej Karpathy – "Let's build GPT"** aur nanoGPT repo
- **Hugging Face NLP Course** – free aur bohot practical
- **Stanford CS224n** – NLP ke liye deep dive

> **Project:** Scratch se chhota GPT likho aur kisi text corpus pe train karo. Urdu poetry, cricket commentary, koi bhi text jo tumhe pasand ho. Colab ke free GPU pe chal jayega.

Phase 6 · LLMs aur Generative Models

8 HAFTE

WORKING WITH LLMS

- Hugging Face ecosystem: `transformers`, `datasets`, `tokenizers`, `peft`
- Full fine-tuning vs parameter efficient fine-tuning
- **LoRA aur QLoRA** – kam GPU memory mein ye life saver hain
- Quantization (4-bit, 8-bit) aur inference optimization
- Instruction tuning aur dataset formatting

RAG AUR APPLICATIONS

- Embeddings aur semantic search
- Vector databases (FAISS, Chroma, Pinecone)
- Chunking strategies, retrieval quality, re-ranking
- Prompt engineering aur systematic evaluation
- Agents aur tool calling ka basic idea

ALIGNMENT AUR GENERATIVE MODELS

- RLHF aur DPO ka conceptual understanding
- Diffusion models, agar image generation mein interest hai
- GANs aur VAEs ka basic idea (ab kam use hote hain lekin concepts useful hain)

> **Project:** Do cheezein banao. Ek, kisi open model ko apne domain data pe LoRA se fine-tune karo. Do, ek RAG system banao jo tumhare apne documents pe kaam kare aur uski accuracy measure karo.

Phase 7 · Deployment aur MLOps

4–6 HAFTE

Ye phase tumhe baqi 90% candidates se alag karta hai. Zyadatar log Jupyter notebook tak ruk jate hain aur wahi unki sabse bari kamzori hoti hai.

- Model ko **FastAPI** mein wrap kar ke REST API banana
- **Docker** basics aur containerization
- Deployment: **Hugging Face Spaces** (free, yahan se shuru karo), phir AWS ya GCP
- Experiment tracking: **Weights & Biases** ya MLflow
- Model versioning aur model registry
- Monitoring, logging, data drift detection
- CI/CD ka basic idea

RESOURCES

- **Full Stack Deep Learning** – free online course, deployment ke liye best
- **Made With ML** – MLOps practices

> **Project:** Apne kisi pichle model ko live deploy karo, public URL ke sath. Ek chhota frontend bhi banao taake koi bhi use kar sake. Resume mein link daalo.

Wo cheezein jo roadmap mein nahi hotin lekin zyada matter karti hain

1. HAR PHASE KE BAAD PROJECT BANAO

Aath courses complete karne se ek deployed project behtar hai. Recruiters certificates nahi dekhte, kaam dekhte hain. GitHub profile tumhara asli resume hai.

2. COURSES COLLECT KARNA BAND KARO

Ye sabse common bimari hai. Log 12 courses enroll karte hain aur ek bhi khatam nahi karte. Ek course lo, poora karo, project banao, phir agla. Aadha chhora hua course zero value rakhta hai.

3. PAPERS PADHNA PHASE 5 SE SHURU KARO

Shuru mein samajh nahi aayenge, ye bilkul normal hai. Abstract, figures aur conclusion se shuru karo, phir method section. Papers With Code achi jagah hai jahan paper ke sath implementation bhi milta hai.

4. KAGGLE COMPETITIONS MEIN HISSA LO

Phase 2 ke baad shuru kar do. Medal na bhi mile, discussion forums aur public notebooks se jo seekhoge wo kisi course mein nahi milega. Top solutions padhne ki aadat daalo.

5. CONSISTENCY ZYADA MATTER KARTI HAI SPEED SE

Rozana do ghante, hafte mein 6 din. Ye weekend pe 12 ghante karne se behtar hai. Deep learning mein concepts ek dusre pe build hote hain, gaps se sab tootta hai.

> **Time estimate, bilkul honest:** Hafte mein 15 se 20 ghante consistently do to 10 se 14 mahine lagenge. Agar koi tumhe "3 mahine mein AI engineer" bech raha hai, wo jhoot bol raha hai.

> **GPU ka masla:** Colab ke free tier se Phase 5 tak aaram se kaam chal jayega. Uske baad Colab Pro ya Kaggle ke weekly GPU hours use karo. Apna GPU tab lena jab bilkul clear ho jaye ke tum seriously continue kar rahe ho, pehle nahi.

Portfolio checklist

Roadmap khatam hone tak tumhare paas ye sab hona chahiye:

☐	Item
☐	GitHub profile with clean README on har project
☐	NumPy se likha hua scratch neural network
☐	Apne data pe train kiya hua image classifier
☐	Scratch se likha hua chhota GPT
☐	Ek fine-tuned LLM (LoRA)
☐	Ek deployed RAG application with public link
☐	Kam se kam do Kaggle competition submissions
☐	Ek technical blog post ya detailed writeup

Ye roadmap ek guide hai, qanoon nahi. Agar kisi phase mein tumhe pehle se cheezein aati hain to skip kar do. Agar kisi phase mein atak jao to ruk kar us par extra waqt do. Sequence important hai, exact duration nahi.